

# OOP LAB 5

## SOLVED

### Walkthrough Task:

```
W T.cpp
1  #include<iostream>
2  using namespace std;
3  class Rectangle{
4  private:
5      double length;
6      double width;
7  public:
8      Rectangle(double l, double w){
9          length = l;
10         width = w;
11     }
12     double area(){
13         return length * width;
14     }
15     ~Rectangle(){
16         cout<<"Destructor Called For Object";
17     }
18 };
19 int main()
20 {
21     Rectangle r1(12.5,15.6);
22     cout<<"Area = "<<r1.area()<<endl;
23 }
```

```
C:\Users\ALAM-PC\Desktop\§
Area = 195
Destructor Called For Object
-----
Process exited after 0.5596 seconds with return value 0
Press any key to continue . . . |
```

### Practice Task 1:

```
W T.cpp  P T1.cpp
1  #include<iostream>
2  using namespace std;
3  class Student{
4  private:
5      string name;
6      int rollnumber;
7  public:
8      Student(string n, int r){
9          name = n;
10         rollnumber = r;
11         cout<<"Parameterized Constructor"<<endl;
12     }
13     ~Student(){
14         cout<<"Destructor Called For Object"<<endl;
15     }
16     void display(){
17         cout<<endl;
18         cout<<"Name: "<<name<<endl;
19         cout<<"Roll Number: "<<rollnumber<<endl;
20         cout<<"\n=====\\n";
21     }
22 };
23 int main()
24 {
25     Student s1("Ali",45);
26     Student s2("Hamza",12);
27     Student s3("Usman",56);
28     Student s4("Umar",22);
29     s1.display();
30     s2.display();
31     s3.display();
32     s4.display();
33     return 0;
34 }
```

```
C:\Users\ALAM-PC\Desktop\§
Parameterized Constructor
Parameterized Constructor
Parameterized Constructor
Parameterized Constructor

Name: Ali
Roll Number: 45

=====

Name: Hamza
Roll Number: 12

=====

Name: Usman
Roll Number: 56

=====

Name: Umar
Roll Number: 22

=====

Destructor Called For Object
Destructor Called For Object
Destructor Called For Object
Destructor Called For Object

-----
Process exited after 0.6193 seconds with return value 0
Press any key to continue . . . |
```

## Practice Task 2:

```
WT.cpp PT1.cpp PT2.cpp
1 #include<iostream>
2 using namespace std;
3 class BankAccount{
4 private:
5     int accountNumber;
6     double balance;
7 public:
8     BankAccount(int a, double b){
9         accountNumber = a;
10        balance = b;
11    }
12    void deposit(double amount){
13        balance = balance + amount;
14        cout<<"Balance After Deposit: "<<balance<<endl;
15    }
16    void withdraw(double amount){
17        if(amount <= balance){
18            balance = balance - amount;
19            cout<<"Balance After Withdraw: "<<balance<<endl;
20        }
21        else{
22            cout<<"Insufficient Balance"<<endl;
23        }
24    }
25    ~BankAccount(){
26        cout<<"Account Closed: "<<accountNumber<<endl;
27    }
28 };
29 int main()
30 {
31     BankAccount b1(123,5000);
32     b1.deposit(2000);
33     b1.withdraw(1000);
34     BankAccount b2(999,1000);
35     b2.deposit(3000);
36     b2.withdraw(1500);
37     BankAccount b3(452,2500);
38     b3.deposit(4000);
39     b3.withdraw(2000);
40     BankAccount b4(253,1000);
41     b4.deposit(1000);
42     b4.withdraw(2000);
43     BankAccount b5(523,1500);
44     b5.deposit(1000);
45     b5.withdraw(3000);
46     return 0;
47 }
```

```
C:\Users\ALAM-PC\Desktop\
Balance After Deposit: 7000
Balance After Withdraw: 6000
Balance After Deposit: 4000
Balance After Withdraw: 2500
Balance After Deposit: 6500
Balance After Withdraw: 4500
Balance After Deposit: 2000
Balance After Withdraw: 0
Balance After Deposit: 2500
Insufficient Balance
Account Closed: 523
Account Closed: 253
Account Closed: 452
Account Closed: 999
Account Closed: 123

-----
Process exited after 0.3135 seconds with return value 0
Press any key to continue . . .
```

## Practice Task 3:

```
WT.cpp PT1.cpp PT2.cpp PT3.cpp
1 #include<iostream>
2 using namespace std;
3 class Product{
4 private:
5     string productName;
6     double price;
7     int quantity;
8 public:
9     Product(string pn, double p, int q){
10        productName = pn;
11        price = p;
12        quantity = q;
13    }
14    void display(){
15        cout<<"Product Name: "<<productName<<endl;
16        cout<<"Product Price: "<<price<<endl;
17        cout<<"Product Quantity: "<<quantity<<endl;
18        cout<<"===== "<<endl;
19    }
20    ~Product(){
21        cout<<"Product "<<productName<<" removed from the inventory"<<endl;
22    }
23 };
24 int main()
25 {
26     Product p1("Laptop",500000,5);
27     p1.display();
28     Product p2("Phone",80000,10);
29     p2.display();
30     Product p3("Watch",5000,25);
31     p3.display();
32     return 0;
33 }
```

```
C:\Users\ALAM-PC\Desktop\
Product Name: Laptop
Product Price: 500000
Product Quantity: 5
=====
Product Name: Phone
Product Price: 80000
Product Quantity: 10
=====
Product Name: Watch
Product Price: 5000
Product Quantity: 25
=====
Product Watch removed from the inventory
Product Phone removed from the inventory
Product Laptop removed from the inventory

-----
Process exited after 0.2887 seconds with return value 0
Press any key to continue . . .
```